UNIVERSITY OF MARYLAND GLOBAL CAMPUS

Adelphi · Syllabus · Other

CMSC 451 7383 Design and Analysis of Computer Algorithms (2262)

CMSC-451

Spring 2026 Section 7383 3 Credits 03/11/2026 to 05/05/2026

Faculty Contact

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Course Description

Prerequisites: CMSC 150 and CMSC 315 (or CMIS 350). A presentation of fundamental techniques for designing and analyzing computer algorithms. The aim is to apply big-O estimates of algorithms and proof-of-correctness techniques and to design algorithms. Basic methods include divide-and-conquer techniques, search and traversal techniques, dynamic programming, greedy methods, and induction. Programming projects are included.

Course Introduction

Different algorithms can often be written to perform the same task. In such cases, some criteria should be used to select one of the competing methods. Usually, the important criteria are space (i.e., memory) requirements and time requirements. Time requirements can be estimated based on a worst-case or an average-case basis.

In this course, we will focus on the analysis of important algorithms used in a wide variety of applications (including sorting, searching, string matching, and graph-theoretic algorithms). There are classes of algorithms for which it is likely that no efficient algorithms exist. In dealing with such problems, it is useful to know in advance that no efficient algorithms exist. We will introduce the theory (NP-completeness) behind these algorithms. We will also include programming projects to empirically compare algorithm performance.

Course Outcomes

After completing this course, you should be able to

- apply Big-O estimates of algorithms, using techniques such as activation trees, recurrence relations, the Little Master Theorem, and summations
- design algorithms, using techniques such as brute force, divide and conquer, and greedy
- compare algorithms experimentally to select the most efficient implementations
- categorize the NP-completeness of problems to determine their fundamental complexity
- apply proof-of-correctness techniques to verify correctness of algorithms

Course Materials

Click to access your course materials information (http://webapps.umgc.edu/UgcmBook/BPage.cfm?C=CMSC%20451&S=7383&Sem=2262)

Class Guidelines

Faculty Information

To locate information within your LEO classroom, log in and review your faculty member's information, which is found in the **Course Home** of your classroom under **Meet your Professor** section.

Contacting your Faculty Member

You can email your faculty member at the email address listed under **Faculty Contact**.

You can also use the Instant Message feature to send a message to your faculty member. Simply click **Classlist** from the top navigation, click the **Faculty** tab, check the box next to your faculty member's name and click **Instant Message**.

Click the Classroom Walkthrough Videos link below for more information on navigating the classroom.

Classroom Walkthrough Videos Link (https://www.umgc.edu/current-students/learning-resources/leo)

Contacting Advising or the Department

- If you have questions related to the course content or any of the graded deliverables, please contact your instructor.
- For questions and concerns related to advising, you can call 800-888-8682 (toll-free), or, write to undergrad.advisor@umgc.edu (if you are an undergraduate student), or grad.advisor@umgc.edu (if you are a graduate student).
- For other questions and concerns about this class, you can contact your program director by writing to infotechnologydept@umgc.edu. Please be sure to mention the course name, course number, and your section number in the "Subject:" field of your email. Your email will be treated confidentially.

Grading Information

This course consists of the following graded items (See Class and Assignment Schedule for details):

8 Weekly Discussions @ 12.5pts each	100
6 Homework @ 50pts each	300
2 Projects @ 300pts each	600
Final Calculated Grade	**1,000**

Definition of Academic Rigor

UMGC defines academic rigor as the degree to which students demonstrate content mastery, application of critical thinking skills and adherence to UMGC's code of academic integrity.

This definition implies three components to academic rigor:

1. Content mastery to include the subject matter of the course as well as mastery of those core curriculum goals established for the course (for example, information literacy, effective writing).
2. Application of critical thinking skills to include the degree to which the student can present and defend original thinking on the subject matter, including synthesis and analysis of key concepts.
3. Academic integrity to include the degree to which student demonstrates academic honesty defined in UMGC's code of academic integrity.

Participation

By registering for a Web-based course, you have made a commitment to participate in course conferences as well as other online activities. Participation for this course is defined as proactive discussion in weekly discussion activities. This requires you to actively reflect on weekly readings and to develop original ideas in your responses. You are expected to demonstrate critical thinking and your understanding of the content in the assigned readings as they relate to the issues identified in the conference discussion.

You are expected to respond to the main discussion topic(s) each week and read and respond to other student posts to contribute additional knowledge to the class. Note that your online conference participation counts significantly toward your final grade. Interacting and responding to discussion items earlier in the week as opposed to the end of the week is encouraged and rewarded.

When communicating with others in this class always work to be respectful.

Policy on Late Submission

Assignments are expected to be submitted on time. Assignments are made available in the beginning of the course so that students have enough lead time in which to prepare, ask questions, and seek help.

A late assignment will be penalized unless a major accident, illness, or deployment (for active-duty military), with supporting documentation, prevents a student from submitting his/her assignment on time. Students should keep their instructor informed in advance about military deployments or other events that may impact their progress and participation in the course and request for a deadline extension.

The standard Late Penalty consists of a **5%** reduction for up to 2 days late, then **15%** for work submitted up to a week late. After a week, late assignments will not be accepted or graded.

Quizzes and discussion posts must be completed on time to receive a grade.

Faculty have the discretion to relax the requirements for required documentation and/or the application of such late penalties if the student circumstances warrant.

The one exception is the end of the course. No assignments will be accepted after the last day of class unless a student has satisfied the requirements for an Incomplete according to UMGC Policy 170.71 Grade of Incomplete. The student must request for an Incomplete before the end of the semester or term and get approval from the instructor.

Extra Credit

There are no extra credit opportunities for the class.

Project Descriptions

Descriptions and details for all homework and projects are listed in the assignments area found in the Navigation bar under assignments and in the content area.

Academic Policies

ACADEMIC INTEGRITY

University of Maryland Global Campus (UMGC) has adopted a Philosophy of Academic Integrity (https://www.umgc.edu/current-students/learning-resources/academic-integrity/philosophy) to guide the university's commitment to a culture of academic integrity. Our approach cultivates socially responsible personal and professional behaviors and traits. All members of the University community must maintain the highest level of integrity across the academic experience.

Resources - UMGC provides an Academic Integrity Tutorial (https://www.umgc.edu/current-students/learning-resources/academic-integrity/tutorial) and an Integrity & Ethics Badge as learning resources. These resources include information on the fundamentals of academic integrity and how to apply your own personal ethics to coursework as a UMGC student and in other settings. Other Academic Integrity resources and guidelines are found at https://www.umgc.edu/current-students/learning-resources/academic-integrity (https://www.umgc.edu/current-students/learning-resources/academic-integrity).

Role of Instructor - Your instructor is your primary resource for how to uphold the highest ethical standards in the context of the specific requirements for this course.

Student Responsibility – You are responsible for using UMGC-approved resources to understand key academic integrity concepts and to support your own academic success through practices that uphold values of integrity: honesty, trust, fairness, respect, responsibility, and courage.

Turnitin - Turnitin is enabled within the classroom to support the development and assessment of authentic student writing. To learn more about Turnitin, the feedback it provides, how to use that feedback to improve your work, and your options regarding the inclusion of your work in the Turnitin database, visit University guides for Turnitin at https://www.umgc.edu/current-students/learning-resources/writing-center/writing-and-turnitin (https://www.umgc.edu/current-students/learning-resources/writing-center/writing-and-turnitin).

ARTIFICIAL INTELLIGENCE

In keeping with our mission to prepare learners for careers and life after college, UMGC embraces the importance of artificial intelligence (AI) as part of that future. The efficient, effective, and ethical use of artificial intelligence tools to assist learning can prepare you for your career, especially tasks involving the top abilities that employers are seeking: problem solving, creativity, critical thinking, collaboration, and analysis. But, while using AI in some stages of your assignments will help to prepare you for your job, your use must also reflect the ethical requirements of your chosen profession and UMGC's Academic Integrity policy.

Within the context of professional ethics and academic integrity, the University generally permits the use of Generative Artificial Intelligence (GenAI) tools, like ChatGPT, for tasks such as generating ideas, brainstorming, finding background information, clarifying research questions, and improving one's grasp of coding or math concepts. These and other uses of artificial intelligence (AI) tools are acceptable as long as they align with an assignment's requirements and its intended learning goals. In addition, any AI content a student submits as part of an assignment should include citation or other forms of attribution. To assist you, student resources on proper use and attribution of AI tools to support learning can be found on UMGC's Library Website at this link: https://libguides.umgc.edu/artificial-intelligence (https://libguides.umgc.edu/artificial-intelligence).

CLASSROOM CIVILITY

University of Maryland Global Campus is committed to the success of our global community and values the diverse identities and backgrounds of our students, faculty, and staff. Each one of us has a broader life and set of experiences beyond UMGC that we bring with us to each interaction. Sharing your story with your classmates provides opportunities to learn, relate, and gain inspiration from each other. Engagement often begins with introductions at the beginning of the course. Sharing your preferred name, preferred pronouns, and other details about yourself and your life builds a foundation for connection, understanding, and a richer and more personalized learning experience.

We also recognize that some of life's responsibilities and challenges outside of the classroom, such as childcare, a change in employment status, or illness, have an impact on success in a course. To the extent you are comfortable, we encourage you to communicate with your faculty member or Success Coach about any concerns you have for this course or as a student at UMGC so we can help you navigate potential obstacles and stay on track to achieve your goals.

Students are expected to work together cooperatively, and treat fellow students and faculty with respect, showing professionalism and courtesy in all interactions. Please review the Code of Civility for more guidance on interacting in UMGC classrooms: https://www.umgc.edu/current-students/student-life-and-

support/student-handbook/civility-code (https://www.umgc.edu/current-students/student-life-and-support/student-handbook/civility-code).

POLICIES AND GUIDELINES

UMGC is committed to creating a climate in which everyone can thrive. UMGC's Non-Discrimination and Anti-Harassment Policy (https://www.umgc.edu/administration/policies-and-reporting/policies/administration-policies/non-discrimination-and-anti-harassment) (Policy VI-1.00) and Sexual Misconduct Policy (https://www.umgc.edu/administration/policies-and-reporting/policies/administration-policies/sexual-misconduct) (Policy 041.00) protect students, faculty, and staff.

Here you will find UMGC's Non-Discrimination Statement (https://www.umgc.edu/terms-conditions/nondiscrimination-statement).

Students with disabilities who need accommodations in a course are encouraged to contact Accessibility Services at accessibilityservices@umgc.edu (mailto:accessibilityservices@umgc.edu), or call 240-684-2287.

The following academic policies and procedures apply to this course and your studies at UMGC.

Policy Number	Policy Title and Description
150.25	Academic Integrity Policy (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/academic-integrity) The University expects all members of the university community—students, faculty, and staff—to use guidelines to work with and promote integrity. If you are aware of any academic misconduct, please contact integrity@umgc.edu. All cases of academic misconduct will be addressed in accordance with Policy 150.25 (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/academic-integrity) and associated procedures. You are expected to engage in new learning that furthers your development of knowledge, skills, and abilities in each course. According to this policy, you may not submit a substantial portion of any coursework that you have submitted to any course previously without express written approval through assignment guidelines or other forms of communication. You must use UMGC course materials responsibly. Uploading course materials to any website outside of UMGC's online classroom is prohibited by this policy.
V-1.03	Code of Student Conduct (https://www.umgc.edu/administration/policies-and-reporting/policies/student-affairs/code-of-student-conduct)

Policy Number	Policy Title and Description
170.40 170.41 170.42	The following policies describe the requirements for the award of each degree: Degree Completion Requirements for Graduate Students (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/graduate-school-degree-completion-requirements) Degree Completion Requirements for a Bachelor's Degree (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/bachelors-degree-completion-requirements) Degree Completion Requirements for an Associate's Degree (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/associates-degree-completion-requirements)
V-1.30	Student Reasonable Accommodation Policy (https://www.umgc.edu/administration/policies-and-reporting/policies/student-affairs/reasonable-accommodations-students) - Students are responsible for self-identifying with Accessibility Services to inform the university about medical conditions and request academic accommodations.
170.71	Policy on Grade of Incomplete (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/grade-of-incomplete-policy) - Additionally, the mark of I depends on the following two criteria: Students who have completed 60% of their coursework with a grade of B or better for graduate courses or C or better for undergraduate courses and students who request an I before the end of the term. The mark of I is not available for noncredit courses.
170.72	Course Withdrawal Policy (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/course-withdrawal) - Students must follow drop and withdrawal procedures and deadlines available at https://www.umgc.edu/admission/academic-calendar (https://www.umgc.edu/admission/academic-calendar) under Academic Calendar.
130.80	Procedures for Review of Alleged Arbitrary and Capricious Grading (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/capricious-grading-review) – Appeals may be made on final course grades as described herein.
IV-3.20	Intellectual Property (https://www.umgc.edu/administration/policies-and-reporting/policies/research/intellectual-property) - All university faculty, staff, and students must comply with University guidelines on the use of copyrighted material. Uploading UMGC or faculty copyrighted material without authorization degrades and corrupts the integrity of the teaching and learning experience and is a potential violation of UMGC policy and copyright law. You must obtain permission to post UMGC or other's copyrighted material to third-party websites, including social learning network sites. UMGC reserves the right to take appropriate action to remove copyrighted material uploaded without authorization.

Policy Number	Policy Title and Description
205.06	Calculation Of Grade-Point Average (GPA) for Inclusion on Transcripts and Transcript Requests (https://www.umgc.edu/administration/policies-and-reporting/policies/academic-affairs/grade-point-average-calculation-for-inclusion-on-transcripts-and-transcript-requests) - Note: Undergraduate and graduate courses have different Grading Policies. See the Grading Policies section of the Course Syllabus.
X-1.12	Acceptable Use (https://www.umgc.edu/administration/policies-and-reporting/policies/info-governance-security-technology/acceptable-use) - The security of the online classroom is critical to ensuring a strong culture of academic integrity and authentic education at the University. It is a violation of the University's policies for anyone to share logon, password, and any other secure information about a UMGC online account, including credentials required to access the online learning environment.

GRADING

According to UMGC's grading policy, the following marks are used:

	Undergraduate	Graduate
A	90-100%	90-100%
B	80-89%	80-89%
C	70-79%	70-79%*
D	60-69%	N/A**
F	59% or below	69% or below
FN	Failure-Non attendance	Failure-Non attendance
G	Grade Pending	Grade Pending
P	Passing	Passing
S	Satisfactory	Satisfactory
U	Unsatisfactory	Unsatisfactory
I	Incomplete	Incomplete
AU	Audit	Audit
W	Withdrew	Withdrew

* The grade of "B" represents the benchmark for graduate courses. Students must maintain a Grade Point Average (GPA) of 3.0 or higher. Classes where final grade of C or F places a student on Academic Probation must be repeated.
** UMGC does not award the grade of D in graduate courses.

GRADE ROUNDING

Scores to individual assignments are calculated based on rubrics in the class and are not rounded to the whole point. The final grade for the course is determined by weighted average and will be rounded to the nearest whole point using mathematical rule (grades with .5 and above to be rounded to the next whole point).

EXTRA CREDIT

Assignments are designed to enable students to achieve course objectives and succeed in the program. In the interest of equity and fairness, there will be no extra credit opportunities. All assignments are identified in the syllabus.

COURSE EVALUATION SURVEY

UMGC values its students' feedback. You will be asked to complete an online evaluation toward the end of the term. The primary purpose of this evaluation process is to assess the effectiveness of classroom instruction in order to provide the best learning experience possible and make continuous improvements to every class. Responses are kept confidential. Please take full advantage of this opportunity to provide your feedback.

LIBRARY SUPPORT

Extensive library resources and services are available online, 24 hours a day, seven days a week at https://libguides.umgc.edu/home (https://libguides.umgc.edu/home) to support you in your studies. The UMGC Library provides research assistance in creating search strategies, selecting relevant databases, and evaluating and citing resources in a variety of formats via its Ask a Librarian service.

TUTORING SUPPORT

UMGC offers tutoring on specific topics (such as writing, reading comprehension, and statistics) to all students in all courses and programs. Additionally, course-specific tutoring is available for select courses. All tutoring is accessed via the classroom, and this is the only means of interacting with UMGC-approved tutors. Tutors enhance learning by explaining concepts and guiding processes; tutors do not complete any work on the student's behalf. Additional information and learning resources can be found at www.umgc.edu/tutoring (http://www.umgc.edu/tutoring) and https://www.umgc.edu/ewc (https://www.umgc.edu/ewc).

EXTERNAL LINK DISCLAIMER

This course may contain links to external sites neither owned nor maintained by UMGC. UMGC bears no responsibility for the accuracy, legality, or content of external sites or for that of subsequent links. In addition, the terms of use, security policies, and privacy policies may differ from those of UMGC. Contact

the external site for answers to questions regarding its content, terms of use, and policies.

LEARNING MANAGEMENT SYSTEM SUPPORT

Those requiring technical assistance can access Help@UMGC Support directly in LEO under the Help menu. Additional technical support is available 24 hours a day, seven days a week via self-help and live chat at https://www.umgc.edu/help (https://www.umgc.edu/help) or by phone toll-free at 888-360-8682.

SYLLABUS CHANGES

All items on this syllabus are subject to change at the discretion of the Instructor and the Office of Academic Affairs.

Class & Assignment Schedule

SESSION	TOPICS	ASSIGNMENTS
Week 1 Mar 11 - Mar 17	• Introduction • Analyzing Algorithms: The 2-d Maxima Problem • Summations and Analyzing Programs with Loops	• Week 1 discussions • Homework 1
Week 2 Mar 18 - Mar 24	• Asymptotics • Divide and Conquer and MergeSort • Recurrences	• Week 2 discussions • Homework 2
Week 3 Mar 25 - Mar 31	• Heaps and HeapSort • HeapSort Analysis and Partitioning • QuickSort • Lower Bounds for Sorting	• Week 3 discussions • Homework 3
Week 4 Apr 1 - Apr 7	• Dynamic Programming: Longest Common Subsequence • Dynamic Programming: Chain Multiplication • Dynamic Programming: Minimum Weight Triangulation	• Week 4 discussions • Project 1

Week 5 Apr 8 - Apr 14	• Graphs: Background and Breadth-First Search • Depth-First Search • Topological Sort and Strong Components	• Week 5 discussions • Homework 4
Week 6 Apr 15 - Apr 21	• Minimum Spanning Trees and Kruskal's Algorithm • Prim's and Baruvka's Algorithms for MSTs • Dijkstra's Algorithm for Shortest Paths	• Week 6 discussions • Homework 5
Week 7 Apr 22 - Apr 28	• All-Pairs Shortest Paths	• Week 7 discussions • Homework 6
Week 8 Apr 29 - May 5	• NP-Completeness: Languages and NP • NP-Completeness: Reductions	• Week 8 discussions • Project 2

Students can access their complete list of assignments and their corresponding due dates within the **Assignments** section of the classroom by navigating to the **Assignments** section of the class from the main navigation bar. Follow the link below, and then click **Assignments**, for a video demonstration on how to utilize this feature.

[Classroom Walkthrough Videos Link (https://www.umgc.edu/current-students/learning-resources/leo)](https://www.umgc.edu/current-students/learning-resources/leo)

Students also have access to a calendar tool on the course homepage within the classroom.